

ZION CESAR I. PANITAN

Laoag City, Ilocos Norte | +63 910 670 3978 | zionpanitan44@gmail.com
<https://github.com/hyper-lab> | <https://hyper-lab.github.io/Portfolio/>

PROJECTS

CasePoint – AI-Driven Semantic Search & Indexing System (Capstone)

2025 – 2026

Backend Developer Lead | Team of 5.

- Led all backend development for an AI-powered document retrieval platform built for Regional Trial Court Branch 17 in Batac City, replacing a fully manual case-file search process with a RAG-based semantic search system deployed over LAN.
- Architected the full retrieval pipeline OCR ingestion via Tesseract, sentence-transformer embedding generation, vector indexing in ChromaDB via LlamaIndex, and semantic query execution with metadata filtering by case type and topic enabling court staff to surface relevant documents in seconds instead of manually searching physical folders.
- Designed and built all REST APIs in FastAPI covering document ingestion, query routing, user session management, and role-based access control (RBAC), ensuring only authorized personnel could access sensitive case records.
- Integrated the FastAPI backend with a React/Vite frontend featuring real-time streaming responses, then coordinated with the team to deliver a fully functional, multi-user system within the semester timeline.
- Presented the completed system to the faculty panel and received commendation for technical execution and real-world applicability; CasePoint was formally accepted as a capstone deliverable for RTC Branch 17.

EXPERIENCE

Freelance Electronics & Computer Technician

2021 – Present

Self-Employed, Laoag City

- Diagnosed and repaired board-level faults on consumer electronics including inverter air conditioners and appliances, applying circuit schematic reading and component-level replacement (IPMs, capacitors, resistors) to restore functionality and save clients the cost of unit replacement.
- Resolved complex PC hardware and software issues including GPU repairs and OS troubleshooting — for individual and small-business clients, building a repeat customer base through reliable, fast turnaround.
- Managed end-to-end client engagements independently: diagnosis, parts sourcing, repair, and post-service follow-up developing strong problem-solving, client communication, and self-management skills alongside technical expertise.

Transformation Team Intern

September 2025 – December 2025

Aretex Business Services Inc., Laoag City

- Mapped and documented end-to-end accounting process flows for Accounts Payable (AP) and Accounts Receivable (AR) workflows, producing standardized procedure references used by the operations team to ensure consistency across client accounts.
- Translated complex daily accounting activities into clear, structured process diagrams, bridging the gap between technical system capabilities and financial workflows reducing ambiguity in cross-team handoffs.
- Collaborated with senior team members to identify bottlenecks in existing workflows, providing data-backed recommendations that contributed to process improvement discussions during the internship period.

Photojournalist2023 – 2025

The Port – Official College Publication, MMSU

- Captured compelling visual content for the official campus publication covering breaking news, academic events, and student life consistently meeting strict editorial deadlines for each publication cycle.
- Applied professional lighting, composition, and timing techniques to produce imagery that elevated the publication's visual quality and reader engagement across print and digital formats.

Freelance Photographer2023 – 2024

Self-Employed

- Directed and executed high-quality photography for weddings and large-scale events, managing lighting, composition, and strict timeline constraints to deliver complete photo sets on deadline.
- Built a client portfolio through referrals by consistently delivering polished, professionally edited outputs that met client expectations for milestone events.

EDUCATION

Bachelor of Science in Information Technology

Graduated: June 11, 2026

Mariano Marcos State University – Main Campus, Batac City, Ilocos Norte

- GWA: 1.740 Special Award
- Capstone Project: CasePoint AI-driven semantic search and indexing system for RTC Branch 17, Batac City (see Projects)
- Relevant Coursework: Data Structures & Algorithms, Database Systems, Web Development, Software Engineering, Machine Learning, Computer Networks

CERTIFICATIONS

- IOT Technologies - Huawei (January 2025)
- 3RD Digital Transformation Forum - Information Technology Center (March 27 2025)
- Certificate of IT Internship Completion - Aretex Business Services Inc. (September – December 2025)
- Data Analytics Essentials - Cisco Academy (October 2024)

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, HTML5, CSS3, SQL, PHP

Frontend: React, Next.js, Tailwind CSS, Vite, REST APIs

Backend: Node.js, FastAPI, MySQL and Laravel

Tools & Platforms: Git, GitHub

Analysis & Documentation: Business Process Mapping, Workflow Analysis, Technical Documentation, Systems Analysis.

Networking: Manual Configuration Cisco Packet Tracer.

Hardware & Electronics: Board-Level Diagnostics, Soldering, Circuit Schematic Reading.

Computer Systems: PC Hardware and Software Troubleshooting.

REFERENCES: Dr. Annjeanette Alain D. Galang | aadgalang@mmsu.edu.ph | Adviser